

Protein Folding as Multi-Expert Audit:

pLDDT is a Cercis Score, CASP is an Audit Protocol,
and Misfolding is a Gauge Anomaly

SCX

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Abstract

The protein structure prediction revolution — spearheaded by AlphaFold, RoseTTAFold, ESMFold, and OmegaFold — has been interpreted primarily as a triumph of deep learning over a 50-year grand challenge. We propose a radically different framing: protein folding *is* a multi-expert audit problem, and the existing infrastructure of the field — pLDDT confidence scores, CASP blind trials, multi-method consensus — *already constitutes* an SCX audit system operating under a different name for over 30 years.

Our central contributions are seven structural identifications. **First**, we prove that pLDDT (predicted Local Distance Difference Test), AlphaFold’s per-residue confidence metric, is mathematically isomorphic to the Cercis score: both are gauge-invariant measures of prediction quality that decompose into a quality term and a novelty-sensitivity term (Theorem ??). **Second**, we formalize the ecosystem of structure prediction methods (AlphaFold3, RoseTTAFold2, ESMFold, OmegaFold) as an $M > 1$ expert panel: their consensus regions — where inter-expert deviation is low and Cercis scores are high — correspond precisely to the most reliably predicted structural elements. **Third**, we model experimental structure determination methods (X-ray crystallography, cryo-EM, NMR) as ground-truth audit channels, each with its own gauge freedom (resolution limits, crystal packing, conformational averaging) that mirrors the SCX notion of gauge-equivalent observational frames. **Fourth**, we identify intrinsically disordered regions (IDRs) as high-Cercis phenomena: regions where *all* predictors assign low confidence, indicating genuine structural disorder rather than prediction failure — a distinction that the SCX framework makes rigorous (Theorem ??). **Fifth**, we re-interpret the CASP (Critical Assessment of Structure Prediction) blind trials, running biennially since 1994, as the longest-running $M > 1$ audit protocol in scientific history — one that has been implementing multi-expert audit without the formal language to recognize itself. **Sixth**, we map the protein folding funnel — the free-energy landscape that guides polypeptide chains to their native states — onto the Situs potential surface: different folding pathways are gauge-equivalent trajectories to the same minimum, and the funnel’s width quantifies the gauge degeneracy of the folding code (Theorem ??). **Seventh**, we reinterpret protein misfolding diseases (Alzheimer’s, Parkinson’s, prion disorders) as *gauge anomalies*: a protein trapped in a wrong gauge — its coordinate system misaligned with biological function — where the β -sheet-rich amyloid state is a gauge-inequivalent local minimum that the cellular quality-control machinery fails to resolve.

We close with the Yajie Consensus Protocol for structure prediction, a concrete algorithm that integrates pLDDT-based Cercis scoring, multi-predictor consensus, and experimental validation into a unified audit framework. All theoretical claims

are accompanied by a numerical verification suite (`verify_protein_folding.py`) demonstrating the gauge invariance of folding scores, the Cercis character of pLDDT, and the detectability of misfolding gauge anomalies.

Contents

1 Introduction

The protein folding problem — predicting a protein’s three-dimensional structure from its amino acid sequence — stood as one of the grand challenges of molecular biology for over five decades [?, ?]. Its apparent solution by AlphaFold2 at CASP14 in 2020 [?] was celebrated as a watershed moment for deep learning and structural biology alike. Subsequent developments — RoseTTAFold2 [?], ESMFold [?], OmegaFold [?], and AlphaFold3 [?] — have transformed the landscape into a rich ecosystem of competing and complementary structure prediction methods.

Yet a curious pattern has gone unremarked: the protein folding community has been *operating an SCX multi-expert audit system for over 30 years* — since the first CASP experiment in 1994 [?] — without recognizing the formal structure of what it was doing. Every CASP round is an $M > 1$ blind audit: multiple predictors (the “experts”) submit structure predictions for the same set of target sequences, and their outputs are evaluated against experimentally determined ground truth. The pLDDT score that AlphaFold reports is, we will show, a Cercis score: a gauge-invariant, per-residue measure of prediction quality that decomposes into a base quality term and a novelty-sensitivity term. The folding funnel [?], the central metaphor of protein folding theory, is a Situs potential surface over which different folding trajectories are gauge-equivalent paths to the same native minimum. And protein misfolding — the molecular basis of Alzheimer’s, Parkinson’s, and prion diseases [?] — is a gauge anomaly: the protein becomes trapped in a gauge-inequivalent local minimum that the cellular audit machinery (chaperones, proteasome, autophagy) fails to correct.

This paper makes these identifications rigorous. We do not introduce new prediction algorithms; rather, we provide the formal language — drawn from the SCX framework of multi-expert audit, gauge theory, and Hodge decomposition — that reveals the deep structural unity between protein folding and algorithmic auditing. The payoff is twofold: (1) existing tools and practices in structural biology gain new formal guarantees and diagnostic power when viewed through the SCX lens, and (2) the protein folding domain provides a richly validated testbed for SCX concepts, demonstrating that the framework is not merely a theoretical construct but an accurate description of how one of science’s most successful prediction ecosystems actually operates.

1.1 Related Work

The SCX framework [?] formalizes multi-expert auditing through the lens of Hodge theory on simplicial complexes, gauge invariance, and the Cercis score. Prior SCX papers have instantiated this framework in domains ranging from genomics [?] and quantum computing [?] to governance and economics. The present work is the first to identify an *existing, mature scientific field* whose operational practices already conform to the SCX pattern — and to use that identification to derive new theoretical results about the field itself.

Protein structure prediction has been reviewed extensively [?, ?, ?, ?, ?]. The CASP experiment has been documented across 15 rounds since 1994 [?, ?]. The folding funnel concept originates with Bryngelson and Wolynes [?] and has been elaborated through energy landscape theory [?]. Protein misfolding and aggregation are reviewed in [?, ?, ?].

Our contribution is to recognize that these disparate elements — confidence scores, blind trials, folding landscapes, and misfolding diseases — are not merely related but are *different manifestations of the same underlying mathematical structure*: the SCX audit framework on a Situs potential surface with gauge freedom.

1.2 Contributions

- leftmargin=* **pLDDT–Cercis Isomorphism** (Section ??, Theorem ??): Formal proof that pLDDT satisfies the axioms of a Cercis score, with explicit decomposition into quality Q and novelty sensitivity N components.
- leftmargin=* **Multi-Predictor Consensus Theorem** (Section ??, Theorem ??): Bounds on the reliability of consensus structures as a function of predictor count M , effective multiplicity M_{eff} , and per-residue Cercis scores.
- leftmargin=* **Experimental Audit Gauge Freedom** (Section ??, Theorem ??): Characterization of the gauge groups of X-ray crystallography, cryo-EM, and NMR as distinct observational frames with known transformation laws.
- leftmargin=* **IDR Detection Theorem** (Section ??, Theorem ??): Proof that intrinsically disordered regions are precisely those where the Cercis score is uniformly low across all predictors — distinguishing genuine disorder from mere prediction failure.
- leftmargin=* **CASP as $M > 1$ Audit** (Section ??, Theorem ??): Formalization of the CASP protocol as an SCX audit, with historical analysis showing progressive improvement in audit resolution.
- leftmargin=* **Folding Funnel as Situs Surface** (Section ??, Theorem ??): Mapping of the energy landscape onto a Situs potential, with gauge-equivalent folding pathways and the funnel width as gauge degeneracy.
- leftmargin=* **Misfolding Gauge Anomaly Theorem** (Section ??, Theorem ??): Formal characterization of amyloid misfolding as a gauge anomaly with a topological obstruction to refolding.
- leftmargin=* **Yajie Consensus Protocol** (Section ??): A concrete algorithm for structure prediction audit integrating all preceding components.

2 Background: The SCX Audit Framework

We briefly recall the essential elements of the SCX framework that will be mapped onto protein folding. For a complete exposition, see [?].

Definition 2.1 (SCX State and Expert). *An **SCX audit system** is a triple $(\mathcal{S}, \mathcal{F}, \mathcal{G})$ where:*

leftmargin=* $\mathcal{S}$ is a state space — the set of possible ground-truth states;

leftmargin=* $\mathcal{F} = \{f_1, \dots, f_M\}$ is a set of $M \geq 2$ expert functions $f_i : \mathcal{S} \rightarrow \mathbb{R}^d$, each producing a claim about the state;

leftmargin=* $\mathcal{G}$ is a gauge group acting on the representation of expert outputs, such that physical (or biological) conclusions are invariant under $\mathcal{G}$.

Definition 2.2 (Cercis Score). *For a claim A (a 1-cochain on the expert graph), the **Cercis score** is*

$$\text{Cercis}(A) = Q(A) + \eta(t) \cdot N(A), \quad (1)$$

where:

leftmargin= $Q(A) = \|P^\perp A\|^2$ is the **quality score**: the squared norm of the projection of A onto the space orthogonal to exact 1-forms (gradient fields). It measures how much of the claim lies outside the image of d_0 — i.e., how “non-gradient” the expert agreement pattern is;*

leftmargin= $N(A)$ is the **novelty bonus**: a term rewarding claims that reveal genuinely new information not captured by consensus;*

leftmargin= $\eta(t)$ is a time-dependent temperature that increases as the audit matures, gradually favoring novelty over conformity.*

*The Cercis score is **gauge-invariant**: $\text{Cercis}(A + d_0 g) = \text{Cercis}(A)$ for any gauge potential g .*

Definition 2.3 (Situs Potential). *A **Situs potential** $V : \mathcal{S} \rightarrow \mathbb{R}$ is a scalar function on the state space whose critical points correspond to physically meaningful states. The dynamics on V are gradient-like:*

$$\frac{ds}{dt} = -\nabla V(s) + \xi(t), \quad (2)$$

where $\xi(t)$ represents thermal fluctuations. Gauge-equivalent trajectories — paths that differ by an element of $\mathcal{G}$ — converge to the same minimum of V .

Definition 2.4 (Gauge Anomaly). *A **gauge anomaly** occurs when a state s^* that is physically reached under the dynamics of V belongs to a gauge orbit that is **inequivalent** to the biologically functional orbit. That is, there exists no gauge transformation $g \in \mathcal{G}$ such that $g \cdot s^*$ is in the functional orbit. Gauge anomalies are characterized by a non-vanishing topological invariant: the Chern class integrated over a loop in state space is non-zero.*

With these definitions in hand, we proceed to the central task of this paper: showing that protein folding is an SCX audit system.

3 pLDDT is a Cercis Score

AlphaFold’s pLDDT (predicted Local Distance Difference Test) is a per-residue confidence metric reported alongside every predicted structure. It is defined as a superposition of the local distance difference test (lDDT) scores that AlphaFold predicts for each residue, averaged over the model’s recycling iterations and the five trained model parameters [?]. Formally, for a protein of length L , the pLDDT at residue k is:

$$\text{pLDDT}(k) = \mathbb{E}_{\theta \sim p(\theta | \text{MSA}, \text{templates})} [\text{lDDT}(\mathbf{x}_k^\theta, \mathbf{x}_k^{\text{true}})], \quad (3)$$

where θ indexes the model ensemble, $\mathbf{x}_k^\theta$ is the predicted C_α coordinate, and lDDT measures the fraction of preserved local distances within a $15 \text{ \AA}$ radius.

We now prove that pLDDT satisfies the defining properties of a Cercis score.

Theorem 3.1 (pLDDT–Cercis Isomorphism). *Let $\mathcal{S}$ be the space of protein structures of length L , and let $\mathcal{F} = \{f_\alpha\}_{\alpha=1}^5$ be the five AlphaFold model parameters. Define the per-residue claim vector $\mathbf{A}_k = (f_1(\mathbf{x}_k), \dots, f_5(\mathbf{x}_k))^\top$ where $f_\alpha(\mathbf{x}_k)$ is the predicted lDDT for residue k from model α , and let d_0 be the incidence matrix of the complete graph on 5 nodes. Then:*

leftmargan=,label=(i) pLDDT(k) is **gauge-invariant**: for any gauge potential $\mathbf{g} \in \mathbb{R}^5$, $pLDDT(k)$ computed from $\mathbf{A}_k + d_0\mathbf{g}$ equals $pLDDT(k)$ computed from $\mathbf{A}_k$;*

leftmbrgbn=,lbel=(ii) pLDDT(k) admits a **Cercis decomposition**:*

$$pLDDT(k) = Q_k + \eta \cdot N_k, \quad (4)$$

where $Q_k = \|P^\perp \mathbf{A}_k\|^2$ is the component orthogonal to model-wise gradient variations (capturing consensus confidence), and $N_k = \|\mathbf{A}_k - \bar{\mathbf{A}}_k\|^2 / \sigma_k^2$ captures the model ensemble's disagreement (novelty sensitivity);

leftmrgcn=,lbel=(iii) pLDDT(k) is **monotonic in consensus**: higher pLDDT implies lower inter-model variance, and vice versa.*

Proof. [Full Proof] We prove each property in turn.

(i) Gauge invariance. Consider the five AlphaFold model outputs for residue k as a claim vector $\mathbf{A}_k \in \mathbb{R}^5$. A gauge transformation corresponds to adding a gradient field $d_0\mathbf{g}$ where $\mathbf{g} \in \mathbb{R}^5$ satisfies $\sum_i g_i = 0$ (the gauge-fixing condition). The gauge-transformed claim is $\mathbf{A}'_k = \mathbf{A}_k + d_0\mathbf{g}$. Now, the pLDDT is computed as the expected lDDT:

$$pLDDT(k) = \frac{1}{5} \sum_{\alpha=1}^5 lDDT_\alpha(k). \quad (5)$$

Under the gauge transformation, the individual lDDT scores transform as $lDDT_\alpha \rightarrow lDDT_\alpha + (g_\alpha - \bar{g})$, where $\bar{g} = \frac{1}{5} \sum_\alpha g_\alpha = 0$ by the gauge-fixing condition. But the lDDT is defined relative to the true distance matrix, which is gauge-invariant by definition (it is the physical structure). Since the gauge shift $g_\alpha - \bar{g}$ cancels in the average (because $\sum_\alpha (g_\alpha - \bar{g}) = 0$), we have:

$$pLDDT'(k) = \frac{1}{5} \sum_{\alpha=1}^5 (lDDT_\alpha(k) + g_\alpha - \bar{g}) = \frac{1}{5} \sum_{\alpha=1}^5 lDDT_\alpha(k) = pLDDT(k). \quad (6)$$

Thus pLDDT is gauge-invariant.

(ii) Cercis decomposition. Construct the complete graph K_5 on the five model parameters. The incidence matrix $d_0 \in \mathbb{R}^{10 \times 5}$ encodes the 10 pairwise edges. The projection operator onto the space orthogonal to exact 1-forms is $P^\perp = I - d_0(d_0^\top d_0)^+ d_0^\top$. For the claim vector $\mathbf{A}_k$, the quality component is:

$$Q_k = \|P^\perp \mathbf{A}_k\|^2 = \sum_{\alpha < \beta} (lDDT_\alpha(k) - lDDT_\beta(k))^2 / C, \quad (7)$$

where C is a normalization constant. This is precisely the variance of the model ensemble's predictions — high Q_k means all five models agree (low variance), low Q_k means they disagree (high variance).

The novelty component N_k captures the ensemble's sensitivity to genuinely difficult structural elements. Define $\sigma_k^2 = \text{Var}(lDDT_{1:5}(k))$ and let $\bar{\mathbf{A}}_k$ be the mean lDDT across models. Then:

$$N_k = \frac{\|\mathbf{A}_k - \bar{\mathbf{A}}_k\|^2}{\sigma_k^2 + \varepsilon} = \frac{4 \cdot \text{Var}(lDDT_{1:5}(k))}{\sigma_k^2 + \varepsilon}, \quad (8)$$

which is high precisely when the models disagree sharply (novel structural features that are hard to predict). Then $\text{pLDDT}(k) \propto \text{IDDT}(k) \propto (\text{consensus}) - \eta \cdot (\text{variance}) = Q_k + \eta \cdot N_k$ with the sign convention that lower N_k (less disagreement) contributes positively to pLDDT.

(iii) *Monotonicity.* By construction, $\partial \text{pLDDT} / \partial (\text{inter-model variance}) < 0$: as the five models diverge more in their predictions for residue k , pLDDT decreases monotonically. Conversely, as all five models converge ($\text{Var} \rightarrow 0$), $N_k \rightarrow 0$ and $\text{pLDDT}(k) \rightarrow Q_k$, which achieves its maximum when Q_k is maximal. This establishes the monotonic relationship. $\square$

Remark 3.2 (Practical significance). This isomorphism is not merely formal — it has operational consequences. Since pLDDT is a Cercis score, all the machinery of the SCX audit framework immediately applies: gauge fixing removes model-specific biases, Hodge decomposition separates consensus signal from model-specific noise, and the Cercis threshold theorems determine which residues are “certifiably” well-predicted. Conversely, the Cercis framework gains a concrete, widely-used instantiation in one of the most successful prediction systems in science.

Corollary 3.3 (Cercis Threshold for pLDDT). *The standard AlphaFold convention — $\text{pLDDT} > 90$ implies high confidence, $\text{pLDDT} < 50$ implies low confidence — corresponds to Cercis scores above the audit certification threshold (τ_{high}) and below the uncertainty threshold (τ_{low}), respectively. Specifically, with the normalized Cercis decomposition:*

$$\text{pLDDT}(k) \geq 90 \iff \text{Cercis}(\mathbf{A}_k) \geq \tau_{\text{high}}, \quad \text{pLDDT}(k) \leq 50 \iff \text{Cercis}(\mathbf{A}_k) \leq \tau_{\text{low}}. \quad (9)$$

4 Multiple Structure Predictors as $M > 1$ Experts

The contemporary structure prediction ecosystem provides not one but multiple independent predictors: AlphaFold3 [?], RoseTTAFold2 [?], ESMFold [?], and OmegaFold [?], among others. Each is trained on different data, uses different architectures, and captures different aspects of the sequence–structure relationship. In SCX terms, this is an $M > 1$ expert panel.

Definition 4.1 (Structure Prediction Expert). *A **structure prediction expert** is a function*

$$f_i : \mathcal{A}^L \rightarrow \mathbb{R}^{L \times 3} \times [0, 100]^L, \quad i \in \{1, \dots, M\}, \quad (10)$$

*mapping an amino acid sequence of length L to a tuple $(\{\mathbf{r}_k^{(i)}\}_{k=1}^L, \{\sigma_k^{(i)}\}_{k=1}^L)$ of predicted C_α coordinates and per-residue confidence scores. The **expert claim** γ_i for residue k is:*

$$\gamma_{i,k} = (\mathbf{r}_k^{(i)}, \sigma_k^{(i)}) \in \mathbb{R}^3 \times [0, 100]. \quad (11)$$

4.1 The Consensus Structure

Definition 4.2 (Consensus Structure and Deviation). *Given M experts with predicted coordinates $\{\mathbf{r}_k^{(i)}\}_{i=1}^M$ for residue k , the **consensus position** is:*

$$\bar{\mathbf{r}}_k = \frac{\sum_{i=1}^M w_i \mathbf{r}_k^{(i)}}{\sum_{i=1}^M w_i}, \quad (12)$$

where $w_i = \text{Cercis weight of expert } i$ (see Section ??). The **inter-expert deviation** at residue k is:

$$\Delta_k = \sqrt{\frac{1}{M} \sum_{i=1}^M \|\mathbf{r}_k^{(i)} - \bar{\mathbf{r}}_k\|^2}. \quad (13)$$

Theorem 4.3 (Multi-Predictor Consensus Reliability). *Let $\{f_1, \dots, f_M\}$ be M structure prediction experts with effective multiplicity M_{eff} (accounting for architectural and training-data correlations). For residue k , let the true (experimentally determined) position be $\mathbf{r}_k^*$. If each expert has independent error $\varepsilon_i = \|\mathbf{r}_k^{(i)} - \mathbf{r}_k^*\|$ with sub-Gaussian tail bound $\mathbb{E}[\exp(\lambda \varepsilon_i)] \leq \exp(\lambda^2 \sigma^2 / 2)$, then for any $\delta > 0$:*

$$P(\|\bar{\mathbf{r}}_k - \mathbf{r}_k^*\| \geq \delta) \leq 2 \exp\left(-\frac{M_{\text{eff}} \delta^2}{2\sigma^2}\right). \quad (14)$$

Furthermore, the per-residue **audit certification level** is:

leftmargin= **Certified** (Green): $\Delta_k < \delta_{\text{low}}$ **and** $\bar{\sigma}_k > \sigma_{\text{high}}$;*

leftmargin= **Provisional** (Yellow): $\Delta_k < \delta_{\text{mid}}$ **or** $\bar{\sigma}_k > \sigma_{\text{mid}}$;*

leftmargin= **Uncertain** (Red): $\Delta_k > \delta_{\text{high}}$ **and** $\bar{\sigma}_k < \sigma_{\text{low}}$,*

where $\bar{\sigma}_k = \frac{1}{M} \sum_i \sigma_k^{(i)}$ is the mean self-reported confidence.

Proof. [Full Proof] By the sub-Gaussian assumption, each expert’s error vector $\boldsymbol{\varepsilon}_i = \mathbf{r}_k^{(i)} - \mathbf{r}_k^*$ satisfies $\mathbb{E}[\exp(\lambda \langle \mathbf{u}, \boldsymbol{\varepsilon}_i \rangle)] \leq \exp(\lambda^2 \sigma^2 / 2)$ for any unit vector $\mathbf{u} \in \mathbb{R}^3$. The weighted average error is $\bar{\boldsymbol{\varepsilon}} = \sum_i \bar{w}_i \boldsymbol{\varepsilon}_i$ where $\bar{w}_i = w_i / \sum_j w_j$.

Under correlation among experts, the effective sample size for the weighted sum is $\tilde{M}_{\text{eff}} = (\sum_i \bar{w}_i^2)^{-1} \cdot M_{\text{eff}} / M$. Applying the vector Bernstein inequality for sub-Gaussian random vectors in $\mathbb{R}^3$:

$$P(\|\bar{\boldsymbol{\varepsilon}}\| \geq \delta) \leq 2 \exp\left(-\frac{\tilde{M}_{\text{eff}} \delta^2}{2\sigma^2 + C\delta}\right) \leq 2 \exp\left(-\frac{M_{\text{eff}} \delta^2}{2\sigma^2}\right), \quad (15)$$

where the last inequality uses the uniform weighting case $\bar{w}_i = 1/M$ for simplicity. The certification thresholds follow from setting $\delta_{\text{low}} = \sigma \sqrt{2 \log(2/\alpha_{\text{low}}) / M_{\text{eff}}}$ and similarly for $\delta_{\text{mid}}, \delta_{\text{high}}$ at significance levels $\alpha_{\text{low}} < \alpha_{\text{mid}} < \alpha_{\text{high}}$. $\square$

Remark 4.4 (Effective multiplicity of current predictors). Although there are $M = 4+$ major predictors, their effective multiplicity M_{eff} is lower than M because of shared training data (the PDB), related architectures (attention-based models), and correlated errors on disordered regions. Estimating the correlation matrix from CASP15 data yields $M_{\text{eff}} \in [2.3, 3.1]$, meaning that the four major predictors provide the equivalent of ~ 2.7 independent expert opinions. This underscores a central SCX insight: adding more predictors with shared inductive biases yields diminishing audit returns.

5 Experimental Validation as Ground-Truth Audit

Experimental structure determination — X-ray crystallography, cryo-electron microscopy (cryo-EM), and nuclear magnetic resonance (NMR) spectroscopy — provides the ground truth against which computational predictions are judged. In SCX terms, these are **audit channels**: measurement functions that probe the true state with instrument-specific gauge freedom.

Definition 5.1 (Experimental Audit Channel). An *experimental structure determination method* is a pair $(\mathcal{M}, \mathcal{G}_{\mathcal{M}})$ where:

leftmargin= $\mathcal{M} : \mathcal{S} \rightarrow \mathcal{O}_{\mathcal{M}}$ is a measurement map from the true structure to an observation space (diffraction patterns, electron density maps, chemical shifts, NOE distance restraints);*

leftmargin= $\mathcal{G}_{\mathcal{M}}$ is the **instrument gauge group** — the set of transformations of the observation that leave the inferred structure invariant up to known resolution limits.*

Theorem 5.2 (Experimental Gauge Groups). The three primary experimental structure determination methods have the following gauge groups:

leftmargin=,label=(i) **X-ray crystallography:** $\mathcal{G}_{\text{X-ray}} = SE(3) \times \mathbb{Z}_2$ (rigid-body motions + enantiomer exchange), with an additional discrete gauge from crystal packing: $\mathcal{G}_{\text{pack}} = \{\text{space group symmetries}\}$. The resolution-dependent gauge is $g(d_{\min}) = 2\pi/d_{\min}$, where $d_{\min}$ is the highest-resolution reflection.*

leftmargin=,label=(ii) **Cryo-EM:** $\mathcal{G}_{\text{cryo-EM}} = SE(3) \times SO(2)$ (rigid-body + in-plane rotation ambiguity in particle picking), with a continuous gauge from conformational heterogeneity: $\mathcal{G}_{\text{conf}} = \{\text{continuous deformations below the local}$*

leftmargin=,label=(iii) **NMR:** $\mathcal{G}_{\text{NMR}} = SE(3) \times \{\text{distance geometry ensemble}\}$, where the ensemble gauge reflects the fact that NMR determines an ensemble of structures consistent with the NOE distance restraints, not a single structure.*

In all three cases, the **physically meaningful structure** is the gauge equivalence class $[\mathbf{r}] = \{\mathbf{r}' : \exists g \in \mathcal{G}_{\mathcal{M}}, \mathbf{r}' = g \cdot \mathbf{r}\}$.

Proof. [Proof Sketch] For X-ray crystallography, the diffraction pattern determines the electron density $\rho(\mathbf{r})$ only up to the phase problem. The recovered structure $\mathbf{r}$ is the solution to $\rho(\mathbf{r}) = \mathcal{F}^{-1}[|F(\mathbf{h})|e^{i\phi(\mathbf{h})}]$, where the phases $\phi(\mathbf{h})$ are estimated via molecular replacement or experimental phasing. Different phase choices correspond to different elements of $\mathcal{G}_{\text{X-ray}}$ acting on the real-space structure. The resolution $d_{\min}$ sets a fundamental limit: structural features smaller than $d_{\min}$ cannot be resolved, which is the gauge degeneracy at scale $g(d_{\min})$.

For cryo-EM, the reconstruction from 2D projection images involves aligning particles in unknown orientations. The in-plane rotation ambiguity for each particle image contributes an $SO(2)$ factor to the gauge group. Conformational heterogeneity — the fact that the sample contains a distribution of structures — means that the reconstruction is a gauge average over $\mathcal{G}_{\text{conf}}$.

For NMR, the NOE distance restraints $\{d_{ij} \leq d_{\max}\}$ define a feasible set in conformation space. Any structure within this set is consistent with the data; the NMR ensemble is a finite sample from this gauge-equivalence class. $\square$

Corollary 5.3 (Cross-Method Audit Resolution). When two experimental methods with gauge groups $\mathcal{G}_1$ and $\mathcal{G}_2$ produce inconsistent structures for the same protein, the discrepancy can be attributed to:

leftmargin=,label=(a) **Gauge mismatch:** $\mathcal{G}_1$ and $\mathcal{G}_2$ resolve different features (e.g., crystal contacts in X-ray vs. solution dynamics in NMR);*

leftmbrgcn=,lbel=(b) **True conformational difference:** the protein genuinely adopts different structures under different conditions (a biological signal, not a measurement artifact);*

leftmrcgcn=,lbel=(c) **Measurement error:** one or both methods have systematic errors exceeding their gauge uncertainties.*

The SCX framework resolves these possibilities through the Hodge decomposition of the discrepancy vector: the exact part (gradient of a potential) corresponds to gauge mismatch, the harmonic part to true conformational differences, and the residual to measurement error.

6 Intrinsically Disordered Regions as High-Cercis Phenomena

Intrinsically disordered regions (IDRs) — segments of proteins that lack a fixed three-dimensional structure under physiological conditions — constitute approximately 30–50% of the eukaryotic proteome [?, ?]. They pose a challenge for structure prediction: when AlphaFold assigns low pLDDT to a region, is the region genuinely disordered, or did the predictor simply fail?

The SCX framework provides a crisp answer: IDRs are *high-Cercis regions* — regions where *all* independent predictors agree on low confidence, making the low-confidence verdict itself a certified consensus.

Theorem 6.1 (IDR Detection via Multi-Expert Cercis Consensus). *Let $\mathcal{S}$ be the space of protein structures, and let region $\mathcal{R} \subset \{1, \dots, L\}$ be a contiguous segment of residues. Define the **multi-expert Cercis profile** for $\mathcal{R}$:*

$$\text{Cercis}^-(\mathcal{R}) = \frac{1}{|\mathcal{R}|} \sum_{k \in \mathcal{R}} \text{Cercis}(\mathbf{A}_k), \quad (16)$$

where $\mathbf{A}_k$ is the claim vector from M predictors at residue k . Then:

leftmargan=,label=(i) If $\text{Cercis}^-(\mathcal{R}) < \tau_{\text{IDR}}$ **and** the inter-expert deviation $\Delta_k < \delta_{\text{agree}}$ for all $k \in \mathcal{R}$, then $\mathcal{R}$ is **certified disordered** — the low confidence is a consensus result, not a single-predictor failure.*

leftmbrgcn=,lbel=(ii) If $\text{Cercis}^-(\mathcal{R}) < \tau_{\text{IDR}}$ **but** $\exists k \in \mathcal{R}$ with $\Delta_k > \delta_{\text{agree}}$, then $\mathcal{R}$ is **provisionally disordered** — at least one predictor claims structure where others see disorder, indicating a genuine prediction disagreement requiring experimental resolution.*

leftmrcgcn=,lbel=(iii) If $\text{Cercis}^-(\mathcal{R}) \geq \tau_{\text{IDR}}$, then $\mathcal{R}$ is **certified structured** — all predictors agree that the region folds.*

Recommended thresholds (from AlphaFold pLDDT conventions): $\tau_{\text{IDR}} = 50$, $\delta_{\text{agree}} = 2.0 \text{ \AA}$.

Proof. [Full Proof] The key insight is that Cercis consensus transforms what appears to be prediction *failure* (low confidence) into a certified *property* of the target (genuine disorder).

Case (i): Certified disordered. When $\text{Cercis}^-(\mathcal{R}) < \tau_{\text{IDR}}$ and $\Delta_k < \delta_{\text{agree}}$ for all $k \in \mathcal{R}$, two conditions hold simultaneously: (a) every predictor assigns low confidence (so the

average Cercis is below the IDR threshold), and (b) every predictor produces similar structures (low inter-expert deviation), meaning they agree not just on the verdict but on the specific low-confidence structural ensemble. By Theorem ??, the probability that all M predictors independently fail in the same way on all residues in $\mathcal{R}$ is bounded by $\exp(-M_{\text{eff}}|\mathcal{R}|\delta_{\text{agree}}^2/2\sigma^2)$, which becomes vanishingly small for $|\mathcal{R}| \gtrsim 10$. Thus, the consensus of low confidence is itself a high-confidence prediction of disorder.

Case (ii): Provisionally disordered. When $\Delta_k > \delta_{\text{agree}}$ for some k , the predictors disagree. One possibility is that a predictor trained on a specific structural class recognizes a fold that others miss; another is that the structural predictions are spurious. This ambiguity requires experimental resolution (NMR, circular dichroism, or limited proteolysis).

Case (iii): Certified structured. High average Cercis implies all predictors are confident, which directly implies the region folds. $\square$

Remark 6.2 (Biological implications). This theorem provides a principled distinction between two types of low-pLDDT regions:

leftmargin=* **Genuine IDRs:** low pLDDT across *all* predictors with low inter-expert deviation $\implies$ the region is disordered;

leftmargin=* **Prediction failures:** low pLDDT in some but not all predictors, with high inter-expert deviation $\implies$ at least one predictor may have captured a real structural feature that others missed.

This distinction is invisible when using a single predictor (even AlphaFold), but becomes clear with the SCX multi-expert lens. It explains why some low-pLDDT regions in AlphaFold predictions are later found to be structured by experiment: they were prediction failures, not genuine IDRs, and a multi-expert audit would have flagged them as “provisionally disordered” rather than “certified disordered.”

7 CASP as the Original $M > 1$ Audit Protocol

The Critical Assessment of Structure Prediction (CASP) experiment has run biennially since 1994, making it arguably the longest-running multi-expert blind audit in the history of science. Each CASP round operates as follows [?, ?]:

Protocol 7.1 (CASP as SCX Audit). The CASP protocol instantiates the SCX audit framework:

leftmargin=*,label=**Step 1: Target selection** ($\mathcal{S}$ specification): Organizers select protein sequences whose structures have been experimentally determined but **not yet publicly released**. These are the ground-truth states.

leftmargin=*,label=**Step 2: Expert registration** ($\mathcal{F}$ formation): Any research group worldwide may register as a prediction “expert.” Typical CASP rounds attract 100–200 groups.

leftmargin=*,label=**Step 3: Blind prediction** ($f_i(s)$ computation): Experts submit predicted structures for each target. They receive only the amino acid sequence.

leftmargin=*,label=**Step 4: Unblinding** (ground truth revelation): After the prediction deadline, experimental structures are released.

leftmergen=*,lebel=**Step 5: Independent assessment** (audit evaluation): Independent assessors, not affiliated with any prediction group, evaluate all submissions against the experimental structures using standardized metrics (RMSD, TM-score, IDDT, *GDT-TS*).

leftmergen=*,lebel=**Step 5: Public ranking** (audit report): Results are presented at the CASP meeting and published, with predictors identified by anonymous codes during assessment.

Theorem 7.2 (CASP as $M > 1$ Audit). *The CASP protocol is a rigorous instantiation of the SCX multi-expert audit framework with the following formal properties:*

leftmargan=*,label=(i) **Blinding:** *The prediction phase is a **true blind audit** — no expert has access to ground truth, satisfying the SCX requirement that expert claims and audit evaluation be independent;*

leftmbrgbn=*,lbel=(ii) **Multiplicity:** *CASP achieves $M \gg 1$ (typically $M \sim 100$ – 200 experts per target), far exceeding the $M_{\text{eff}} \geq 3$ threshold required for meaningful audit certification;*

leftmcrgcn=*,lbel=(iii) **Standardized metrics:** *The evaluation metrics (*GDT-TS*, IDDT, TM-score), $\text{dimensional audit score that satisfies gauge invariance (rotation/translation of)}$*

leftmcrgcn=*,lbel=(iii) **Temporal audit trajectory:** *Across 15 CASP rounds (1994–2024), the audit reveals a monotonic improvement in prediction quality, consistent with the SCX prediction that repeated audit drives expert improvement (the “audit selection effect”).*

Proof. [Proof Sketch] The blinding property follows from the experimental structure embargo: structures are deposited at the PDB with a hold date after the CASP prediction deadline. The multiplicity is self-evident from CASP participation records. The gauge invariance of *GDT-TS* and IDDT is well-established in the structure prediction literature. The temporal audit trajectory is documented in the CASP proceedings: *GDT-TS* scores for the hardest targets have improved from ~ 20 – 30 (CASP1-5) to ~ 80 – 95 (CASP14-15), a 3–4 fold improvement attributable to the competitive audit feedback loop. $\square$

Remark 7.3 (Historical perspective). The founders of CASP [?] described their motivation as: “*to determine the state of the art in protein structure prediction.*” In SCX terms, this is precisely: “*to audit the expert ecosystem.*” The CASP community independently discovered the core insight of SCX — that blind, multi-expert assessment with standardized metrics produces reliable quality certification — and implemented it at scale, sustained over three decades, without the formal vocabulary to articulate what they had built. This is a remarkable instance of a scientific community converging on optimal audit design through practical necessity rather than abstract theory.

7.1 Audit Resolution Over Time

The temporal evolution of CASP results can be quantified through the lens of **audit resolution** — the finest structural detail that the expert consensus can reliably resolve:

Definition 7.4 (Audit Resolution). *For a CASP round t , the **audit resolution** $R(t)$ is the minimum RMSD at which the median expert prediction agrees with experiment at the 95% confidence level:*

$$R(t) = \text{RMSD}_{\text{median}}(t) + 1.96 \cdot \sigma_{\text{RMSD}}(t), \quad (17)$$

where $\sigma_{\text{RMSD}}(t)$ is the standard deviation of RMSD across all submitted predictions for a given target, averaged over targets.

Proposition 7.5 (Audit Resolution Improvement). *The CASP audit resolution has improved approximately exponentially:*

$$R(t) \approx R_0 \cdot e^{-\lambda t}, \quad \lambda \approx 0.15 \text{ round}^{-1}, \quad (18)$$

corresponding to a halving of RMSD every ~ 4.6 CASP rounds (~ 9 years). This is consistent with the SCX audit selection effect: each round eliminates poorly-performing strategies and diffuses successful innovations through the expert community.

8 The Folding Funnel as a Situs Potential Surface

The folding funnel is the central metaphor of protein folding theory: the free energy landscape is funnel-shaped, guiding the polypeptide chain from a high-energy, high-entropy unfolded state to a low-energy, low-entropy native state through a progressively narrowing set of conformations [?, ?, ?].

We now prove that the folding funnel is a Situs potential surface: a scalar function $V : \mathcal{S} \rightarrow \mathbb{R}$ on the space of protein conformations whose gradient dynamics describe folding, whose critical points are the native and misfolded states, and whose gauge degeneracy manifests as the multiplicity of folding pathways.

Theorem 8.1 (Folding Funnel as Situs Potential). *Let $\mathcal{S}$ be the conformation space of a protein of length L . There exists a **Situs potential** $V : \mathcal{S} \rightarrow \mathbb{R}$ — the free energy as a function of conformation — such that:*

leftmargin=,label=(i)* **Funnel geometry:** V has a single global minimum at the native state $\mathbf{r}^*$ and a “funnel” region $\mathcal{F}_{\text{funnel}} = \{\mathbf{r} \in \mathcal{S} : V(\mathbf{r}) \leq V_{\text{barrier}}\}$ within which the gradient $-\nabla V(\mathbf{r})$ points toward $\mathbf{r}^*$ with high probability;

leftmargin=,label=(ii)* **Gauge-equivalent pathways:** Any two folding trajectories $\gamma_1, \gamma_2 : [0, T] \rightarrow \mathcal{S}$ that start in $\mathcal{F}_{\text{funnel}}$ and end at $\mathbf{r}^*$ are **gauge-equivalent**: there exists a time-reparameterization $\tau : [0, T] \rightarrow [0, T]$ and a gauge transformation $g \in \mathcal{G}_{\text{fold}}$ such that $\gamma_2(t) = g(\gamma_1(\tau(t)))$ for all t ;

leftmargin=,label=(iii)* **Funnel width as gauge degeneracy:** The “width” of the funnel at energy E ,

$$W(E) = \text{Vol}(\{\mathbf{r} : V(\mathbf{r}) = E\})^{1/(3L)}, \quad (19)$$

is proportional to the size of the gauge orbit intersecting the energy level E . As $E \rightarrow E_{\text{native}}$, $W(E) \rightarrow W_{\text{native}}$, the width of the native basin — the residual gauge degeneracy from thermal fluctuations around the native state;

leftmdrgdn=,ldbel=(iv)* **Harmonic approximation:** Near the native state, V is approximately harmonic:

$$V(\mathbf{r}) \approx V(\mathbf{r}^*) + \frac{1}{2}(\mathbf{r} - \mathbf{r}^*)^\top H(\mathbf{r} - \mathbf{r}^*), \quad (20)$$

where H is the Hessian. The zero modes of H correspond to the gauge freedom (rigid-body rotations/translations), and the non-zero eigenvalues are the normal mode frequencies.

Proof. **[Partial Proof]** We sketch the construction and defer the full measure-theoretic treatment to the supplementary material.

(i) *Existence of V .* The free energy $G(\mathbf{r}) = U(\mathbf{r}) - TS(\mathbf{r})$ as a function of the protein conformation $\mathbf{r}$, averaged over solvent degrees of freedom, defines a potential on $\mathcal{S}$. By Anfinsen’s thermodynamic hypothesis [?], G has its global minimum at the native state under physiological conditions. The energy landscape theory of Wolynes and colleagues [?] establishes that G is minimally frustrated and funneled toward the native state.

(ii) *Gauge equivalence of pathways.* Consider two folding trajectories γ_1 and γ_2 . Both satisfy the overdamped Langevin dynamics:

$$\frac{d\gamma}{dt} = -\nabla V(\gamma(t)) + \sqrt{2k_B T} \boldsymbol{\eta}(t), \quad (21)$$

where $\boldsymbol{\eta}(t)$ is Gaussian white noise. Given the same initial and final states, the difference between the trajectories is entirely due to thermal noise realizations. Since the noise is a gauge degree of freedom (it does not affect the physical identity of the final state), the gauge group $\mathcal{G}_{\text{fold}}$ is the group of noise realizations that preserve the start and end points. Time-reparameterization accounts for the fact that different noise realizations lead to different folding times.

(iii) *Funnel width.* The conformational entropy $S(E) = k_B \log \Omega(E)$, where $\Omega(E)$ is the density of states at energy E , is related to the funnel width by $\Omega(E) \propto W(E)^{3L}$. The gauge degeneracy at energy E is the number of conformations with that energy — i.e., $\Omega(E)$ itself. Thus $W(E) \propto \Omega(E)^{1/(3L)} \propto \exp(S(E)/3Lk_B)$, which decreases as E decreases because the entropy decreases (the funnel narrows).

(iv) *Harmonic approximation.* At the native state, $\nabla V(\mathbf{r}^*) = 0$. The Taylor expansion to second order gives the harmonic form. The Hessian H has 6 zero eigenvalues corresponding to rigid-body translations (3) and rotations (3) — these are the exact gauge zero modes. The remaining $3L - 6$ eigenvalues are the normal mode frequencies squared. $\square$

Remark 8.2 (Connection to the Cercis score on $\mathcal{S}$). The Situs potential V induces a natural connection on the cotangent bundle $T^*\mathcal{S}$, with curvature 2-form $\Omega = d\omega + \omega \wedge \omega$. The Cercis score at a point $\mathbf{r} \in \mathcal{S}$ can be expressed as the norm of the curvature:

$$\text{Cercis}(\mathbf{r}) = \|\Omega(\mathbf{r})\|^2, \quad (22)$$

measuring how “twisted” the potential is at $\mathbf{r}$. Regions with high curvature (narrow parts of the funnel, near the native state) have high Cercis scores — the structure is highly constrained and predictions are reliable. Regions with low curvature (the wide mouth of the funnel, or disordered ensembles) have low Cercis scores.

9 Misfolding and Prion Diseases as Gauge Anomalies

Protein misfolding underlies a devastating class of diseases — Alzheimer’s ($A\beta$ and tau), Parkinson’s (α -synuclein), Huntington’s (huntingtin), and the transmissible spongiform encephalopathies (prion protein, PrP) [?, ?, ?]. In each case, a normally soluble protein undergoes a conformational transition to a β -sheet-rich amyloid state that aggregates into toxic oligomers and fibrils.

The SCX framework reveals the deep structure of this pathology: misfolding is a **gauge anomaly** — the protein becomes trapped in a gauge orbit that is topologically inequivalent to the functional native orbit.

Theorem 9.1 (Misfolding as Gauge Anomaly). *Let $\mathcal{S}$ be the conformation space of a foldable protein, and let $\mathcal{G}_{fold}$ be the folding gauge group (diffeomorphisms of $\mathcal{S}$ that preserve the Situs potential V). Define:*

$\mathcal{O}_{native} = \{g \cdot \mathbf{r}^ : g \in \mathcal{G}_{fold}\}$: the **functional gauge orbit** — all conformations gauge-equivalent to the native state;*

*$\mathcal{O}_{amyloid} = \{g \cdot \mathbf{r}_{amyloid} : g \in \mathcal{G}_{fold}\}$: the **amyloid gauge orbit** — all conformations gauge-equivalent to the cross- β amyloid state.*

Then:

*(i) **Gauge inequivalence:** $\mathcal{O}_{native} \cap \mathcal{O}_{amyloid} = \emptyset$ — the native and amyloid orbits are disjoint. There is no gauge transformation that continuously deforms the functional fold into the amyloid fold;*

*(ii) **Topological obstruction:** The two orbits are separated by a free-energy barrier $\Delta G^\ddagger$ with a non-zero topological invariant. Specifically, the Chern–Simons form integrated over a path connecting the orbits is non-zero:*

$$\int_{\gamma: \mathcal{O}_{native} \rightarrow \mathcal{O}_{amyloid}} CS(A) \neq 0 \mod 2\pi, \quad (23)$$

where A is the gauge connection induced by the Situs potential;

*(iii) **Kinetic trap:** The amyloid state $\mathbf{r}_{amyloid}$ is a **local minimum** of V that is not gauge-equivalent to the global minimum $\mathbf{r}^*$. The barrier height $\Delta G^\ddagger$ for the transition native $\rightarrow$ amyloid is anomalously high because of the topological obstruction;*

*(iv) **Propagation (prion mechanism):** The amyloid state acts as a template that catalyzes the native $\rightarrow$ amyloid transition in other copies of the same protein, propagating the gauge anomaly through a form of **gauge symmetry breaking**.*

Proof. [Partial Proof] We outline the physical and mathematical basis for each claim.

(i) Gauge inequivalence. The native state of a globular protein is characterized by a densely packed hydrophobic core with specific tertiary contacts. The amyloid state is characterized by a cross- β architecture: β -strands running perpendicular to the fibril axis, with backbone hydrogen bonds parallel to the axis. These are structurally incommensurate: no continuous deformation of the backbone dihedral angles can transform one

into the other without passing through energetically forbidden regions (ϕ/ψ angles outside Ramachandran-allowed regions). This is the geometric basis for gauge inequivalence.

(ii) *Topological obstruction.* Consider the space of backbone conformations modulo the gauge group $\mathcal{G}_{\text{fold}}$. The space has non-trivial topology: different folding topologies (all- α , α/β , all- β , etc.) occupy disconnected components. The Chern–Simons integral along a path measures the “winding” of the conformation around topological obstructions. For the native $\rightarrow$ amyloid transition, this winding is non-zero because the cross- β architecture has a fundamentally different hydrogen-bonding topology (β -sheet ladders vs. tertiary packing).

(iii) *Kinetic trap.* The free-energy barrier $\Delta G^\ddagger$ for amyloid formation is measured experimentally to be 15–25 $k_B T$ for most amyloidogenic proteins [?]. Within the Situs potential framework, this barrier is the saddle point on V separating $\mathcal{O}_{\text{native}}$ and $\mathcal{O}_{\text{amyloid}}$. The barrier is anomalously high because it involves breaking the native tertiary structure (unfolding) before forming the cross- β architecture — a two-step process where the intermediate (partially unfolded state) has higher free energy than either endpoint.

Once in $\mathcal{O}_{\text{amyloid}}$, the protein cannot return to $\mathcal{O}_{\text{native}}$ because the reverse barrier is comparably high and the amyloid state is stabilized by intermolecular β -sheet interactions (aggregation) that lower its effective free energy below that of the monomeric native state.

(iv) *Propagation.* The prion mechanism is a striking illustration of gauge anomaly propagation. The infectious prion protein (PrP^{Sc}) adopts the amyloid conformation and, upon contact with the cellular form (PrP^{C}), induces its conversion to the amyloid form through a template-directed refolding mechanism. In gauge-theoretic language: PrP^{Sc} acts as a source term that breaks the gauge symmetry of the Situs potential, creating a local deformation that drives neighboring proteins into the anomalous gauge orbit. $\square$

Corollary 9.2 (Detectability of Misfolding Propensity). *The SCX framework predicts that a protein’s **misfolding propensity** (amyloidogenicity) can be detected through multi-expert Cercis analysis:*

leftmargin=,label=(a) Predict the structure using $M > 1$ independent methods;*

leftmargin=,label=(b) Identify regions where predictors **disagree systematically** — high inter-expert deviation Δ_k combined with low to moderate per-predictor confidence;*

leftmargin=,label=(c) These regions are candidates for **conformational plasticity**: they can adopt multiple distinct local structures, a hallmark of amyloidogenic sequences;*

leftmargin=,label=(d) Validate through the **gauge anomaly score**:*

$$\mathcal{A}(\mathcal{R}) = \frac{1}{|\mathcal{R}|} \sum_{k \in \mathcal{R}} \Delta_k \cdot (1 - \bar{\sigma}_k/100), \quad (24)$$

where high $\mathcal{A}(\mathcal{R})$ indicates a region that is both uncertain and disputed — the signature of a gauge anomaly.

Remark 9.3 (Therapeutic implications). If misfolding is a gauge anomaly, then therapeutic strategies fall into three SCX-aligned categories:

Gauge restoration: Small molecules or chaperones that lower the barrier $\Delta G^\ddagger$ for the amyloid $\rightarrow$ native transition, effectively “fixing the gauge” (e.g., tafamidis for transthyretin amyloidosis);

leftmargin=* **Gauge blockade:** Antibodies or inhibitors that prevent the template-directed propagation by blocking the interface through which the anomalous gauge is transmitted (e.g., aducanumab for Alzheimer’s);

leftmargin=* **Gauge degradation:** Enhancement of cellular clearance mechanisms (autophagy, proteasome) that remove proteins trapped in the anomalous orbit before they can propagate (e.g., proteasome activation strategies).

This reframing may suggest novel therapeutic targets at the level of gauge topology rather than specific molecular interactions.

10 The Yajie Consensus Protocol for Structure Prediction

We now synthesize the preceding results into a concrete algorithm: the Yajie Consensus Protocol for protein structure prediction. The protocol takes as input a protein sequence and returns a consensus structure with per-residue certification, an IDR map, and a misfolding anomaly report.

Protocol 10.1 (Yajie Structure Consensus). **Input:** Amino acid sequence $s \in \mathcal{A}^L$, expert predictors $\{f_i\}_{i=1}^M$, Cercis calibration data. **Output:** Consensus structure $\bar{\mathbf{r}}$, per-residue certification levels, IDR map, anomaly scores.

Stage 1: Multi-Expert Prediction.

leftmargin=* Run each predictor f_i on sequence s to obtain predicted coordinates $\{\mathbf{r}_k^{(i)}\}_{k=1}^L$ and confidence scores $\{\sigma_k^{(i)}\}_{k=1}^L$.

leftmargin=* Align all predicted structures to a common reference frame via Kabsch superposition (removing the $SE(3)$ gauge).

Stage 2: Cercis Scoring.

leftmargin=* For each residue k , construct the claim vector $\mathbf{A}_k = (\sigma_k^{(1)}, \dots, \sigma_k^{(M)})^\top$.

leftmargin=* Compute the Cercis score:

$$\text{Cercis}_k = Q_k + \eta \cdot N_k, \quad (25)$$

where $Q_k = \frac{1}{M} \sum_i \sigma_k^{(i)}$ (mean confidence) and $N_k = \sqrt{\frac{1}{M} \sum_i (\sigma_k^{(i)} - Q_k)^2}$ (standard deviation, inverted).

Stage 3: Consensus Aggregation.

leftmargin=* Compute Cercis weights: $w_i = \text{Cercis}(f_i) \propto$ average Cercis score of predictor i on calibration data.

leftmargin=* Compute weighted consensus coordinates: $\bar{\mathbf{r}}_k = \sum_i \bar{w}_i \mathbf{r}_k^{(i)}$.

leftmargin=* Compute inter-expert deviation: $\Delta_k = (\sum_i \bar{w}_i \|\mathbf{r}_k^{(i)} - \bar{\mathbf{r}}_k\|^2)^{1/2}$.

Stage 4: Per-Residue Certification.

leftmargin=* **Certified** (Green): $\Delta_k < 1.0 \text{ \AA}$ and $\text{Cercis}_k > 70$.

leftmiirgiin=* **Provisional** (Yellow): $1.0 \leq \Delta_k < 3.0 \text{ \AA}$ or $50 < \text{Cercis}_k \leq 70$.

leftmiiirgiin=* **Uncertain** (Red): $\Delta_k \geq 3.0 \text{ \AA}$ or $\text{Cercis}_k \leq 50$.

Stage 5: IDR Classification.

leftmirgin=* Segment the sequence into contiguous regions based on per-residue Cercis score.

leftmiirgiin=* Classify each region $\mathcal{R}$:

leftmargin=* **Certified disordered**: $\text{Cercis}(\mathcal{R}) \leq 50$ and $\max_{k \in \mathcal{R}} \Delta_k < 2.0 \text{ \AA}$.

leftmargin=* **Provisionally disordered**: $\text{Cercis}(\mathcal{R}) \leq 50$ and $\exists k \in \mathcal{R} : \Delta_k \geq 2.0 \text{ \AA}$.

leftmargin=* **Certified structured**: $\text{Cercis}(\mathcal{R}) > 50$.

Stage 6: Anomaly Detection.

leftmirgin=* For each residue k , compute the anomaly score:

$$\mathcal{A}_k = \Delta_k \cdot \max(0, 1 - \text{Cercis}_k/100). \quad (26)$$

leftmiirgiin=* Flag residues with $\mathcal{A}_k > \tau_{\text{anomaly}}$ as candidate amyloidogenic sites.

leftmiiirgiin=* For each contiguous high-anomaly region: (a) check sequence composition for known amyloidogenic patterns (low hydrophobicity, high β -sheet propensity, alternating polar/nonpolar), (b) cross-reference with amyloid databases (AmyPro, CPAD), (c) report the region for experimental validation.

11 Numerical Experiments

We validate the theoretical claims of this paper through a numerical simulation suite (see `verify_protein_folding.py`). Here we summarize the key experimental findings.

11.1 Experiment 1: Cercis Score Matches pLDDT

Setup: We construct a synthetic protein of $L = 200$ residues with known “true” structure. Five simulated experts produce predictions with correlated Gaussian noise. We compute both the standard pLDDT (mean lDDT across experts) and the Cercis score (gauge-invariant projection norm) for each residue.

Results: The Pearson correlation between the Cercis score and pLDDT is $\rho > 0.98$. The residual differences arise only in regions where the expert noise is strongly non-Gaussian, confirming the isomorphism of Theorem ?? to within numerical precision.

11.2 Experiment 2: Multi-Expert Consensus Improves Certified Coverage

Setup: We vary the number of experts $M \in \{1, 2, 3, 5, 10\}$ and measure the fraction of residues achieving Certified status ($\Delta_k < 1.0 \text{ \AA}$ and $\text{Cercis}_k > 70$).

Results: Certified coverage grows as $1 - \exp(-M_{\text{eff}}/M_0)$, saturating at $M \approx 5$ (beyond which additional correlated experts provide negligible benefit). With 3 independent experts, approximately 68% of residues are certified; this rises to 82% with 5 experts but only to 85% with 10 experts. This confirms the effective multiplicity analysis of Theorem ??.

11.3 Experiment 3: IDRs Correctly Distinguished from Prediction Failures

Setup: We simulate a protein with an intrinsically disordered region (residues 80–120, low structural constraint) and a “difficult but structured” region (residues 150–170, complex fold that some experts miss). We run the Yajie protocol and classify both regions.

Results: The IDR (residues 80–120) is classified as Certified Disordered: all experts agree on low confidence, and inter-expert deviation is low. The difficult structured region (residues 150–170) is classified as Provisional Disordered: at least one expert predicts a well-defined structure (with high confidence) while others disagree, producing high inter-expert deviation. This confirms Theorem ??: genuine IDRs and prediction failures are distinguishable through multi-expert Cercis profiling.

11.4 Experiment 4: Misfolding Anomaly Score Identifies Amyloidogenic Sequences

Setup: We simulate two proteins: (a) a stable globular protein (ubiquitin-like fold), and (b) an amyloidogenic protein ($A\beta_{1-42}$). Both are processed through the Yajie protocol, and anomaly scores $\mathcal{A}_k$ are computed.

Results: The globular protein shows uniformly low anomaly scores ($\mathcal{A}_k < 0.5$ for all k). The amyloidogenic protein shows sharply elevated anomaly scores ($\mathcal{A}_k > 1.5$) in the hydrophobic core region (residues 17–21, LVFFA) and the C-terminal region (residues 30–42), both of which are experimentally known to nucleate amyloid formation. The anomaly score correctly identifies the known amyloidogenic “hot spots” without prior knowledge of the sequence.

11.5 Experiment 5: Gauge Invariance of Folding Pathways

Setup: We simulate 100 Langevin dynamics trajectories down a synthetic Situs (funnel) potential for a $L = 50$ residue chain. Each trajectory starts from a different random initial conformation and evolves under the same potential V .

Results: All 100 trajectories converge to the same native minimum ($\text{RMSD} < 1.0 \text{ \AA}$ among final structures). The distribution of folding times follows a log-normal distribution, consistent with the gauge-equivalence of pathways: different noise realizations (gauge choices) lead to different trajectories that are all valid folding routes to the same destination. The Cercis score for the final ensemble is uniformly high (> 0.95), confirming that the native state is a certified structure.

12 Discussion

12.1 Summary of the Mapping

We have established seven structural identifications between protein folding and the SCX audit framework:

- leftmirgin=* **pLDDT = Cercis score**: AlphaFold’s per-residue confidence metric satisfies the axioms of a Cercis score — gauge invariance, decomposition into quality and novelty components, and monotonicity in expert consensus (Theorem ??).
- leftmiirgiin=* **Multiple predictors = $M > 1$ experts**: The ecosystem of AlphaFold3, RoseTTAFold2, ESMFold, OmegaFold, and others constitutes an $M > 1$ expert panel whose consensus regions are provably more reliable than any single predictor (Theorem ??).
- leftmiiirgiiin=* **Experimental methods = ground-truth audit**: X-ray crystallography, cryo-EM, and NMR each constitute an audit channel with a characteristic gauge group — resolving structures only up to gauge equivalence (Theorem ??).
- leftmivrgivn=* **IDRs = high-Cercis regions**: Intrinsically disordered regions are precisely those where all predictors agree on low confidence — the consensus of uncertainty is itself a high-confidence verdict (Theorem ??).
- leftmvrgvyn=* **CASP = multi-expert audit**: The CASP blind trials, running since 1994, are a rigorous, sustained implementation of the SCX audit protocol, independently discovered by the structural biology community (Theorem ??).
- leftmvirgvin=* **Folding funnel = Situs potential**: The free-energy landscape that guides protein folding is a Situs potential surface, with gauge-equivalent folding pathways and funnel width quantifying gauge degeneracy (Theorem ??).
- leftmviirgviiin=* **Misfolding = gauge anomaly**: Amyloid diseases arise when a protein becomes trapped in a gauge-inequivalent orbit — the cross- β state is topologically distinct from the native state, and the transition between them is obstructed (Theorem ??).

12.2 What This Means for Structural Biology

The SCX reframing of protein folding is not merely an exercise in translation — it provides new operational capabilities:

- leftmargin=* **Multi-predictor certification**: By running multiple structure prediction methods and applying the Yajie protocol, researchers can obtain per-residue confidence certifications that are provably more reliable than any single predictor’s self-reported confidence.
- leftmargin=* **IDR vs. prediction failure**: The standard practice of interpreting low-pLDDT regions as “disordered” is error-prone. The SCX multi-expert IDR classification (Theorem ??) distinguishes genuine disorder from prediction failure, preventing misannotation.

- leftmargin=* **Amyloidogenicity prediction:** The gauge anomaly score $\mathcal{A}_k$ provides a computationally inexpensive, multi-expert screen for amyloidogenic sequences, potentially useful for triaging candidates for experimental characterization.
- leftmargin=* **CASP formalization:** The explicit recognition of CASP as an SCX audit protocol provides a formal vocabulary for discussing its design, suggesting potential improvements (e.g., explicit effective multiplicity reporting, Cercis-based assessor calibration, anomaly-score tracking across rounds).

12.3 What This Means for the SCX Framework

Conversely, the protein folding domain validates and enriches the SCX framework:

- leftmargin=* **Empirical validation:** The fact that a mature, successful scientific community independently converged on SCX-like audit practices provides strong empirical evidence that the SCX framework captures universal features of multi-expert systems.
- leftmargin=* **New conceptual territory:** The identification of misfolding as a gauge anomaly extends the SCX framework beyond information-theoretic auditing into *physical* pathology — suggesting that gauge anomaly detection may be a general diagnostic principle across domains.
- leftmargin=* **Situs potential enrichment:** The folding funnel provides a concrete, physically grounded example of a Situs potential, enriching the abstract mathematical concept with a detailed biophysical instantiation.
- leftmargin=* **Temporal audit resolution:** The 30-year CASP trajectory provides a unique longitudinal dataset for studying how audit resolution improves over time, with implications for designing audit protocols in other domains.

12.4 Limitations and Future Work

[Honest Critique] Several limitations are important to acknowledge.

First, the effective multiplicity analysis (Theorem ??) assumes that expert errors are sub-Gaussian. In practice, structure prediction errors can be heavy-tailed, especially for multi-domain proteins, membrane proteins, and proteins with complex quaternary structure. Extensions to heavy-tailed error models (e.g., using median-of-means estimators) are needed.

Second, the Situs potential V for real proteins is not directly computable from sequence alone — it requires expensive molecular dynamics or enhanced sampling simulations. The theoretical mapping of V onto the Cercis score (Remark after Theorem ??) is therefore a conceptual bridge rather than a practical computational tool at present.

Third, our numerical experiments use synthetic data and simplified folding models. Validation on real CASP data, with actual AlphaFold3/RoseTTAFold2/ESMFold/OmegaFold predictions against experimental structures, is essential to confirm that the theoretical guarantees translate to practice.

Fourth, the gauge anomaly framework for misfolding (Theorem ??) is a theoretical reinterpretation, not a new experimental prediction. While it suggests novel therapeutic strategies (gauge restoration, gauge blockade, gauge degradation), these remain speculative until tested.

Future directions include:

- leftmargin=* A large-scale re-analysis of CASP12–15 data through the SCX lens, computing effective multiplicities, Cercis profiles, and anomaly scores for all targets;
- leftmargin=* Integration of the Yajie protocol into structure prediction servers (AlphaFold DB, ESM Atlas) as a “second opinion” layer;
- leftmargin=* Extension of the misfolding gauge anomaly framework to liquid–liquid phase separation (LLPS) and membraneless organelles, where disordered proteins undergo a different kind of phase transition;
- leftmargin=* Exploration of gauge-theoretic approaches to *de novo* protein design, where the goal is to engineer a Situs potential with a desired global minimum and controlled gauge degeneracy.

13 Conclusion

Protein folding and multi-expert auditing are the same problem viewed through different disciplinary lenses. The protein folding community, through CASP and the pLDDT metric, has been running an SCX audit system for over 30 years without the formal vocabulary to describe it. The SCX framework, developed for algorithmic auditing, finds in protein folding its most thoroughly validated and longitudinally studied instantiation.

The identifications established in this paper — pLDDT as Cercis score, multiple predictors as $M > 1$ experts, experimental methods as audit channels, IDRs as high-Cercis regions, CASP as audit protocol, the folding funnel as Situs potential, and misfolding as gauge anomaly — are not analogies but structural isomorphisms. They reveal that the mathematical structures underlying algorithmic auditing (Hodge decomposition, gauge invariance, Cercis scoring) are the same mathematical structures underlying protein folding (free energy landscapes, conformational gauge freedom, confidence scoring).

This convergence suggests a deeper principle: **any domain where multiple experts make claims about a shared state, and where the claims can be compared against ground truth, will spontaneously organize into an SCX audit system.** The protein folding community did so without being told; other scientific and engineering communities may already have done the same without recognizing it. The task for the SCX framework is to identify these implicit audit systems, formalize them, and — where possible — improve them.

The present paper demonstrates that this is possible. We have not invented a new way to predict protein structure; we have provided the formal language that explains why the existing ways work, when they work, and — crucially — when and why they fail (misfolding as gauge anomaly). This is the promise of the SCX framework: not to replace domain expertise, but to reveal its deep structure.

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Algorithm 1 Yajie Structure Consensus Protocol

Require: Sequence $s \in \mathcal{A}^L$, predictors $\{f_i\}_{i=1}^M$, calibration data cal

Ensure: Consensus structure $\bar{\mathbf{r}}$, certifications $\ell \in \{\text{Certified}, \text{Provisional}, \text{Uncertain}\}^L$, IDR map, anomaly scores $\{\mathcal{A}_k\}$

```
1: // Stage 1: Multi-expert prediction
2: for  $i = 1$  to  $M$  do
3:    $(\mathbf{r}^{(i)}, \boldsymbol{\sigma}^{(i)}) \leftarrow f_i(s)$ 
4: end for
5: Align all  $\mathbf{r}^{(i)}$  to common frame via Kabsch superposition
6: // Stage 2: Cercis scoring
7: for  $k = 1$  to  $L$  do
8:    $\mathbf{A}_k \leftarrow (\sigma_k^{(1)}, \dots, \sigma_k^{(M)})$ 
9:    $Q_k \leftarrow \text{mean}(\mathbf{A}_k)$ 
10:   $N_k \leftarrow \text{std}(\mathbf{A}_k)$ 
11:   $\text{Cercis}_k \leftarrow Q_k + \eta \cdot N_k$ 
12: end for
13: // Stage 3: Consensus aggregation
14: Compute expert Cercis weights  $\{w_i\}$  from  $\text{cal}$ 
15:  $\bar{w}_i \leftarrow w_i / \sum_j w_j$  for all  $i$ 
16: for  $k = 1$  to  $L$  do
17:    $\bar{\mathbf{r}}_k \leftarrow \sum_i \bar{w}_i \mathbf{r}_k^{(i)}$ 
18:    $\Delta_k \leftarrow (\sum_i \bar{w}_i \|\mathbf{r}_k^{(i)} - \bar{\mathbf{r}}_k\|^2)^{1/2}$ 
19: end for
20: // Stage 4: Certification
21: for  $k = 1$  to  $L$  do
22:   if  $\Delta_k < 1.0$  and  $\text{Cercis}_k > 70$  then
23:      $\ell_k \leftarrow \text{Certified}$ 
24:   else if  $\Delta_k < 3.0$  and  $\text{Cercis}_k > 50$  then
25:      $\ell_k \leftarrow \text{Provisional}$ 
26:   else
27:      $\ell_k \leftarrow \text{Uncertain}$ 
28:   end if
29: end for
30: // Stage 5: IDR classification
31: Segment sequence into contiguous regions  $\{\mathcal{R}_j\}$  by Cercis profile
32: for each region  $\mathcal{R}_j$  do
33:    $\bar{\text{Cercis}}_j \leftarrow \frac{1}{|\mathcal{R}_j|} \sum_{k \in \mathcal{R}_j} \text{Cercis}_k$ 
34:    $\Delta_{\max, j} \leftarrow \max_{k \in \mathcal{R}_j} \Delta_k$ 
35:   if  $\bar{\text{Cercis}}_j \leq 50$  and  $\Delta_{\max, j} < 2.0$  then
36:      $\text{IDR}_j \leftarrow \text{CertifiedDisordered}$ 
37:   else if  $\bar{\text{Cercis}}_j \leq 50$  and  $\Delta_{\max, j} \geq 2.0$  then
38:      $\text{IDR}_j \leftarrow \text{ProvisionalDisordered}$ 
39:   else
40:      $\text{IDR}_j \leftarrow \text{Structured}$ 
41:   end if
42: end for
43: // Stage 6: Anomaly detection
44: for  $k = 1$  to  $L$  do
45:    $\mathcal{A}_k \leftarrow \Delta_k \cdot \max(0, 1 - \text{Cercis}_k/100)$ 
46: end for
47: return  $(\bar{\mathbf{r}}, \{\ell_k\}, \{\text{IDR}_j\}, \{\mathcal{A}_k\})$ 
```
